Python Automata Library (PAL) Cheat Sheet

Full details: API Guide · API Reference

import PythonAutomataLibrary as pal · Safe mode is default. Call **pal.FastMode**() before constructing any PAL object. Put substantive model kernels under **@pal.njit(cache=True)**.

Factories and utilities

FastMode() — Select PAL fast mode before constructing...
Seed(seed) — Seed PAL's shared random-number stream;...
Random() — Draw a uniform random value from PAL's...
RandInt(max) — Draw an integer from PAL's random stream...
AwaitWindows(timeout=30) — Wait for PAL windows that are still...
ColorScale(colors, value) — Map a scalar value to an RGB color by...
NewIList() — Create an empty PAL integer list usable...
MooreHood(dim, excludeCenter=False) — Build relative integer offsets for a...
VonNeumannHood(dim, excludeCenter=False) — Build relative integer offsets for a Von...
CircleHood(dim, rad, excludeCenter=False) — Build relative integer offsets within...
NewMultinomial(other=None) — Create a PAL multinomial/binomial...
NewGrid(dimensions, dtype) — Create a typed lattice grid with the...
NewAgentGrid(dimensions, numAgentProps=0, isStackable=False) — Create a grid for individual agents;...
NewPopGrid(dimensions, capacity=None) — Create a lattice of integer population...
NewPDEgrid(dimensions) — Create a lattice-backed scalar field for...

Multinomial

Binomial(n: int, p: float) — Draw a binomial sample with n trials and...
Setup(n: int) — Configure the sampler for n trials...
Sample(p: float) — Draw a binomial sample using the...

IList

__len__() — Return the number of integers currently...
__getitem__(index: int) — Return the integer stored at the given...
Append(i: int) — Append integer i to the end of the list.
Clear() — Remove all values from the list.
Random() — Return a randomly selected value from...
Shuffle() — Shuffle the list in place using PAL's...
All() — Return a detached NumPy array containing...
Iter() — Iterate over the current list values...

AgentGrid

__len__() — Return the number of living agents in...
xDim() — Number of lattice sites along x.
yDim() — Number of lattice sites along y; zero...
zDim() — Number of lattice sites along z; zero...
nDims() — Number of spatial dimensions.
wrapX() — Whether x uses periodic boundary wrapping.
wrapY() — Whether y uses periodic boundary wrapping.
wrapZ() — Whether z uses periodic boundary wrapping.
Box(x1: int, x2: int, y1: int=None, y2: int=None, z1: int=None,...) — Iterate lattice coordinates in the...
Hood(hood, x: int, y: int=None, z: int=None, *, unroll: bool=False) — Apply relative neighborhood offsets at a...
GetPop() — Return the number of living agents in...
Alive(agent: int) — Return whether agent currently...
NewAgentSQ(x: int, y: int=-1, z: int=-1) — Create an agent at the center of a...
NewAgent(x: float, y: float=-1.0, z: float=-1.0) — Create an agent at the supplied...
Dispose(agent: int) — Remove a living agent from the grid; its...
I(agent: int) — Return the linear lattice-site index...
XSQ(agent: int) — Return the agent's integer x site...
YSQ(agent: int) — Return the agent's integer y site...
ZSQ(agent: int) — Return the agent's integer z site...
X(agent: int) — Return the agent's continuous x coordinate.
Y(agent: int) — Return the agent's continuous y coordinate.
Z(agent: int) — Return the agent's continuous z coordinate.
__getitem__(key) — Read a supported per-agent property value.
__setitem__(key, value: float) — Set a supported per-agent property value...

AgentGrid

Tol(x: int, y: int=-1, z: int=-1) — Convert lattice coordinates to the...
ItoX(i: int) — Convert a linear site index to its x...
ItoY(i: int) — Convert a linear site index to its y...
ItoZ(i: int) — Convert a linear site index to its z...
InWrapSQX(value: int) — Wrap an integer x site coordinate when x...
InWrapSQY(value: int) — Wrap an integer y site coordinate when y...
InWrapSQZ(value: int) — Wrap an integer z site coordinate when z...
InWrapX(value: float) — Wrap a continuous x coordinate when x is...
InWrapY(value: float) — Wrap a continuous y coordinate when y is...
InWrapZ(value: float) — Wrap a continuous z coordinate when z is...
DispWrapX(x1: float, x2: float) — Return the shortest signed x...
DispWrapY(y1: float, y2: float) — Return the shortest signed y...
DispWrapZ(z1: float, z2: float) — Return the shortest signed z...
counts() — Read-only occupancy counts for lattice...
LastAgent(x: int, y: int=-1, z: int=-1) — Return the last agent handle at the...
MoveSQ(agent: int, x: int, y: int=-1, z: int=-1) — Move an agent to a lattice site...
Move(agent: int, x: float, y: float=-1.0, z: float=-1.0) — Move an agent to continuous coordinates;...
AgentsAt(x: int, y: int=-1, z: int=-1) — Return the agent handle or handles...
AgentsInRadius(rad: float, x: float, y: float=-1.0, z: float=-1.0, *,...) — Return agents within radius rad of a...
All(shuffle: bool=False) — Return a snapshot of living agent...

Grid

__len__() — Return the number of lattice sites.
__getitem__(key) — Read the value or state selected by the...
__setitem__(key, value) — Assign the value or state selected by...
xDim() — Number of lattice sites along x.
yDim() — Number of lattice sites along y; zero...
zDim() — Number of lattice sites along z; zero...
nDims() — Number of spatial dimensions.
wrapX() — Whether x uses periodic boundary wrapping.
wrapY() — Whether y uses periodic boundary wrapping.
wrapZ() — Whether z uses periodic boundary wrapping.
Box(x1: int, x2: int, y1: int=None, y2: int=None, z1: int=None,...) — Iterate lattice coordinates in the...
Hood(hood, x: int, y: int=None, z: int=None, *, unroll: bool=False) — Apply relative neighborhood offsets at a...
Tol(x: int, y: int=-1, z: int=-1) — Convert lattice coordinates to a linear...
ItoX(i: int) — Convert a linear site index to its x...
ItoY(i: int) — Convert a linear site index to its y...
ItoZ(i: int) — Convert a linear site index to its z...
InWrapX(value: int) — Wrap an x coordinate when x is periodic;...
InWrapY(value: int) — Wrap a y coordinate when y is periodic;...
InWrapZ(value: int) — Wrap a z coordinate when z is periodic;...

PopGrid

__len__() — Return the number of lattice sites.
xDim() — Number of lattice sites along x.
yDim() — Number of lattice sites along y; zero...
zDim() — Number of lattice sites along z; zero...
nDims() — Number of spatial dimensions.
wrapX() — Whether x uses periodic boundary wrapping.
wrapY() — Whether y uses periodic boundary wrapping.
wrapZ() — Whether z uses periodic boundary wrapping.
Box(x1: int, x2: int, y1: int=None, y2: int=None, z1: int=None,...) — Iterate lattice coordinates in the...
Hood(hood, x: int, y: int=None, z: int=None, *, unroll: bool=False) — Apply relative neighborhood offsets at a...
Tol(x: int, y: int=-1, z: int=-1) — Convert lattice coordinates to a linear...
ItoX(i: int) — Convert a linear site index to its x...
ItoY(i: int) — Convert a linear site index to its y...
ItoZ(i: int) — Convert a linear site index to its z...
InWrapX(value: int) — Wrap an x coordinate when x is periodic;...
InWrapY(value: int) — Wrap a y coordinate when y is periodic;...
InWrapZ(value: int) — Wrap a z coordinate when z is periodic;...
__getitem__(key) — Return the integer population count at...